

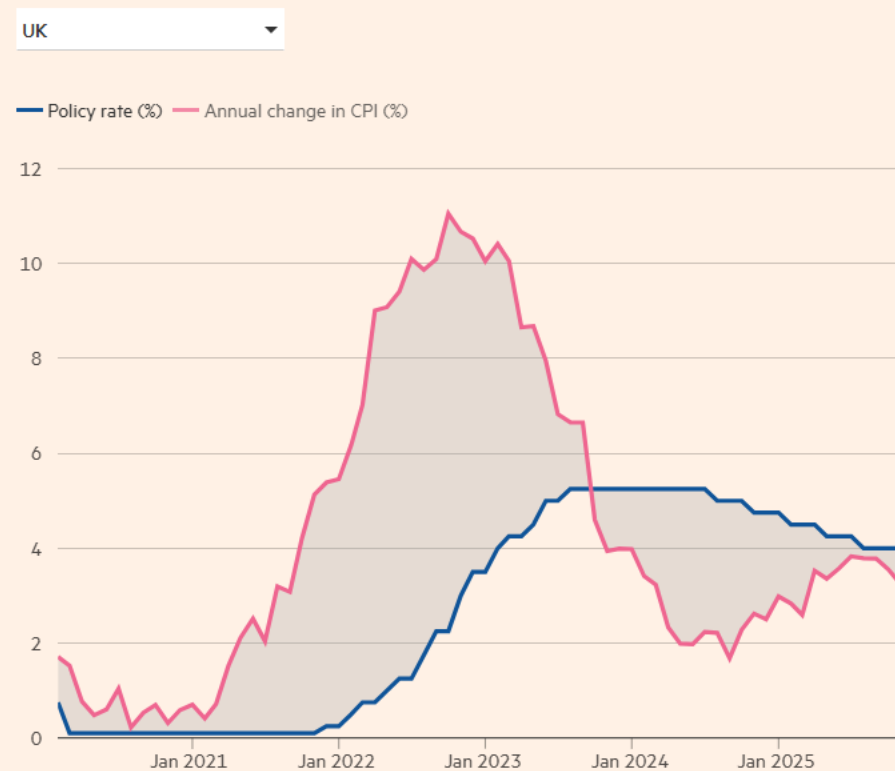
Presentation by

Dr Dalia ElEdel

Week 2

(UMED8H-30-1)

Interest Rates



Source: National statistics offices, central banks via LSEG • Policy rates for Eurozone countries are ECB rate

FINANCIAL TIMES

Main objectives today:

- Discussion of what interest rates are
- Examination of the various types of interest rates: short- and long-term rates, nominal and real rates.
- Explanation of the theory on the determination of interest rates.
- Can changes in interest rates explain disturbances in the stock market?
- An understanding of what interest rate yield curves show (in Bloomberg).

Lecture on Interest Rates

Readings: Burton, M, Nesiba, R., and Brown, B. 2015. An introduction to Financial markets and Institutions. Second Edition. Routledge, Taylor & Francis Group. London & New York, chapter 6

Introduction

- **Interest rates** affect individuals' decisions about whether to spend more or save to buy a house. They also affect decisions by businessmen about whether to invest in new plants or buy Treasury bonds.
- Mainly we want (1) explain the patterns and common threads that link the various interest rates together and (2) to identify the factors that explain the differences.

1-What are Interest Rates?

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- Rental price for money.
- Penalty to borrowers for consuming before earning.
- Reward to savers for postponing consumption.
- Expressed in terms of annual rates.
- As with any price, interest rates serve to allocate resources.

Recap: Measuring Interest Rates

- Present Value:
- A Pound paid to you one year from now is less valuable than a Pound paid to you today
- Why?
 - A Pound deposited today can earn interest and become $£1 \times (1+i)$ one year from today.

Discounting the Future

Let $i = .10$

In one year $\$100 \times (1 + 0.10) = \110

In two years $\$110 \times (1 + 0.10) = \121

or $100 \times (1 + 0.10)^2$

In three years $\$121 \times (1 + 0.10) = \133

or $100 \times (1 + 0.10)^3$

In n years

$\$100 \times (1 + i)^n$

Simple Present Value

PV = today's (present) value

CF = future cash flow (payment)

i = the interest rate

$$PV = \frac{CF}{(1 + i)^n}$$

2-Why do interest rates exist?

2-Why do interest rates exist?

- Because lenders are not altruistic.
- To overcome lenders' time preference.
- Because of inflation
- To compensate lenders for risk

3- Price Expectations and Interest Rates

- Unanticipated inflation benefits borrowers at expense of lenders.
- Lenders charge added interest to offset anticipated decreases in purchasing power.
- Expected inflation is embodied in nominal interest rates: The Fisher Effect.

Fisher Equation

$$i = i_r + \pi^e$$

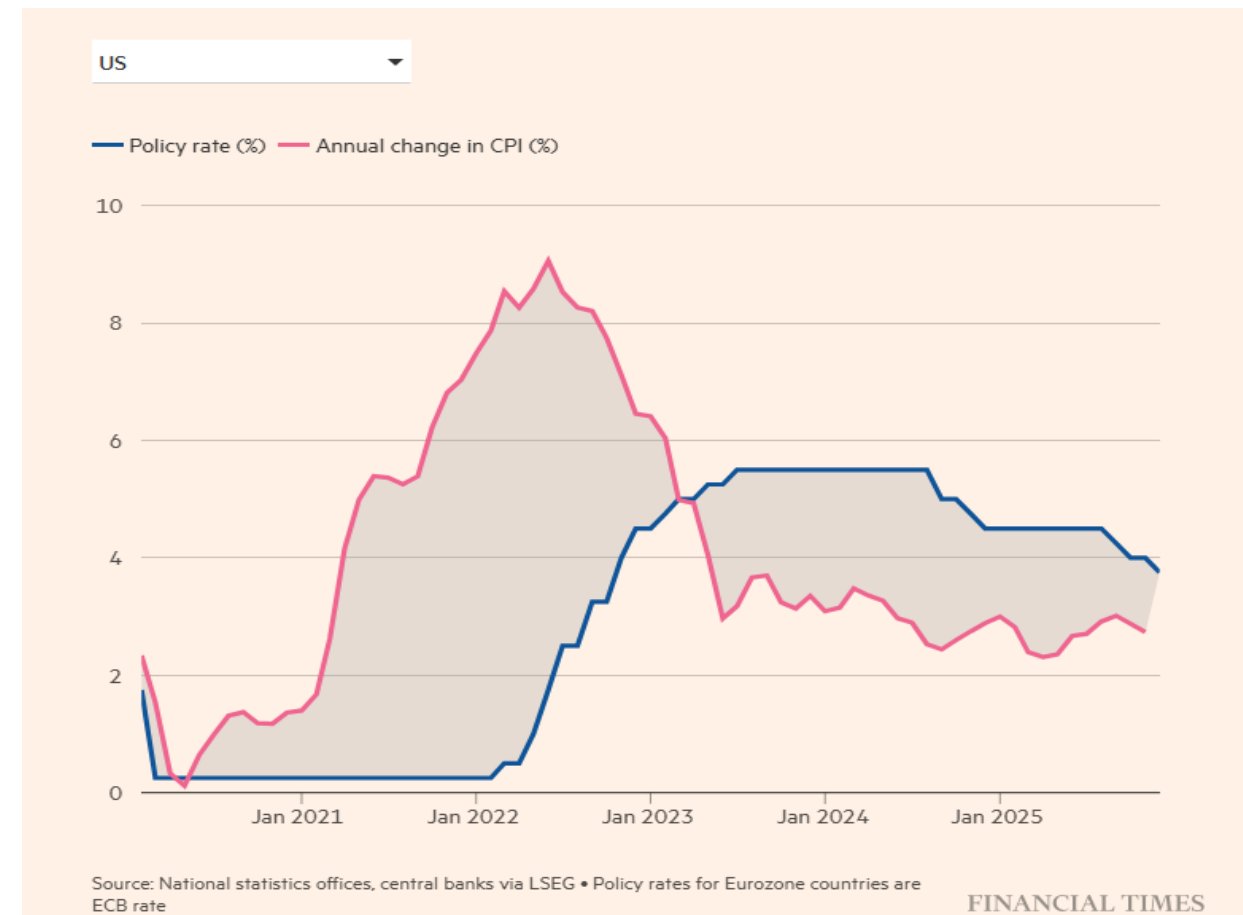
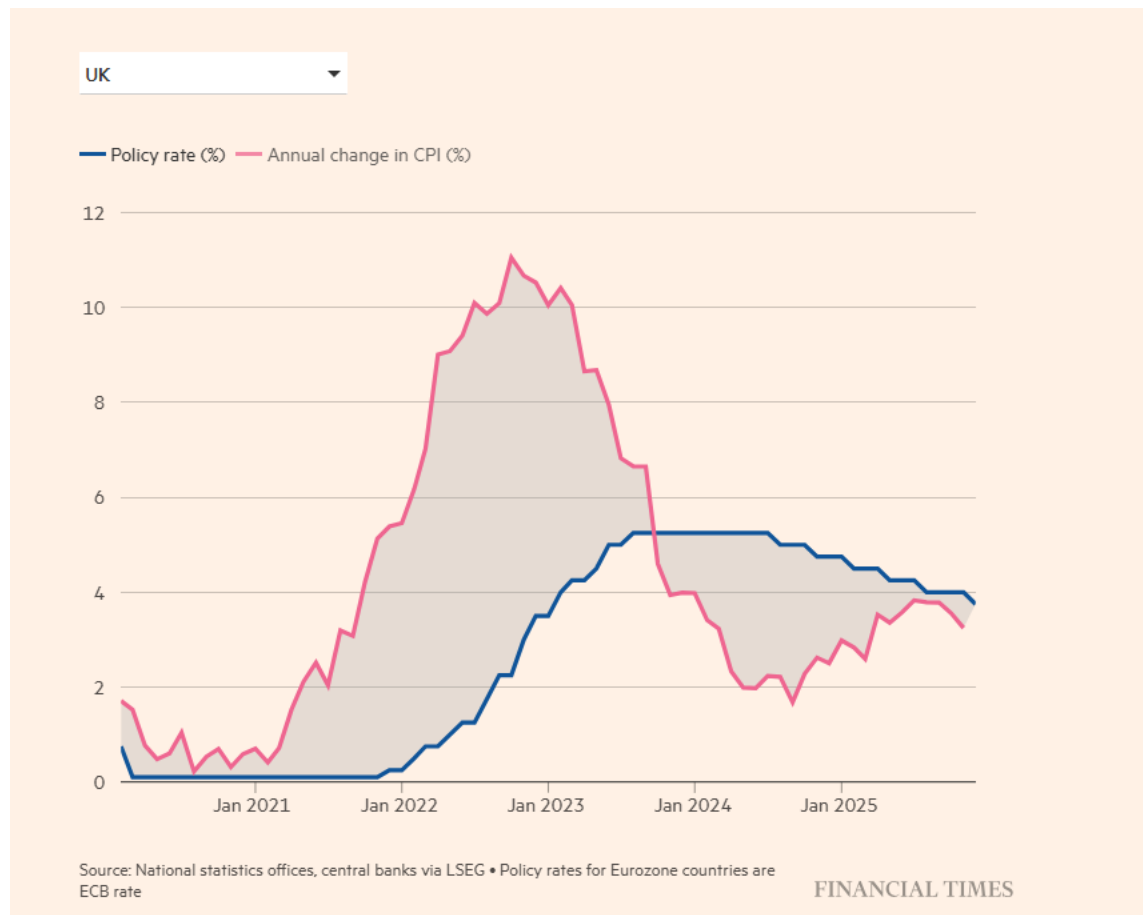
i = nominal interest rate

i_r = real interest rate

π^e = expected inflation rate

When the real interest rate is low,
there are greater incentives to borrow and fewer incentives to lend.

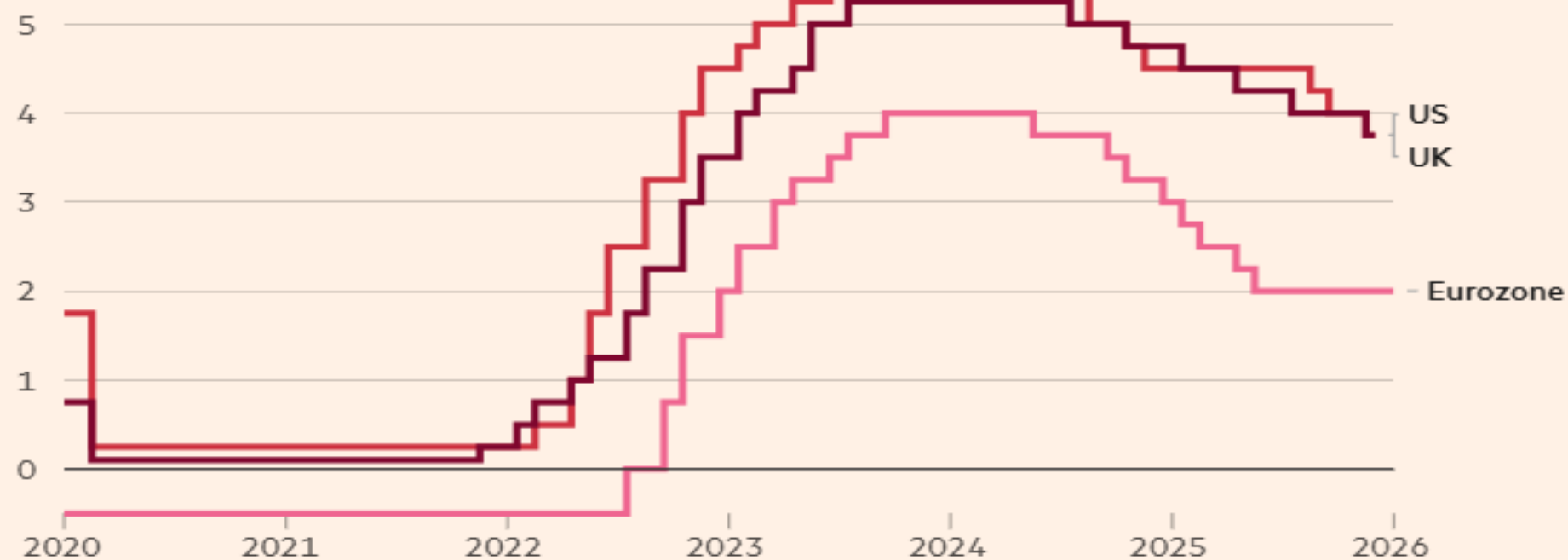
The real interest rate is a better indicator of the incentives to
borrow and lend.



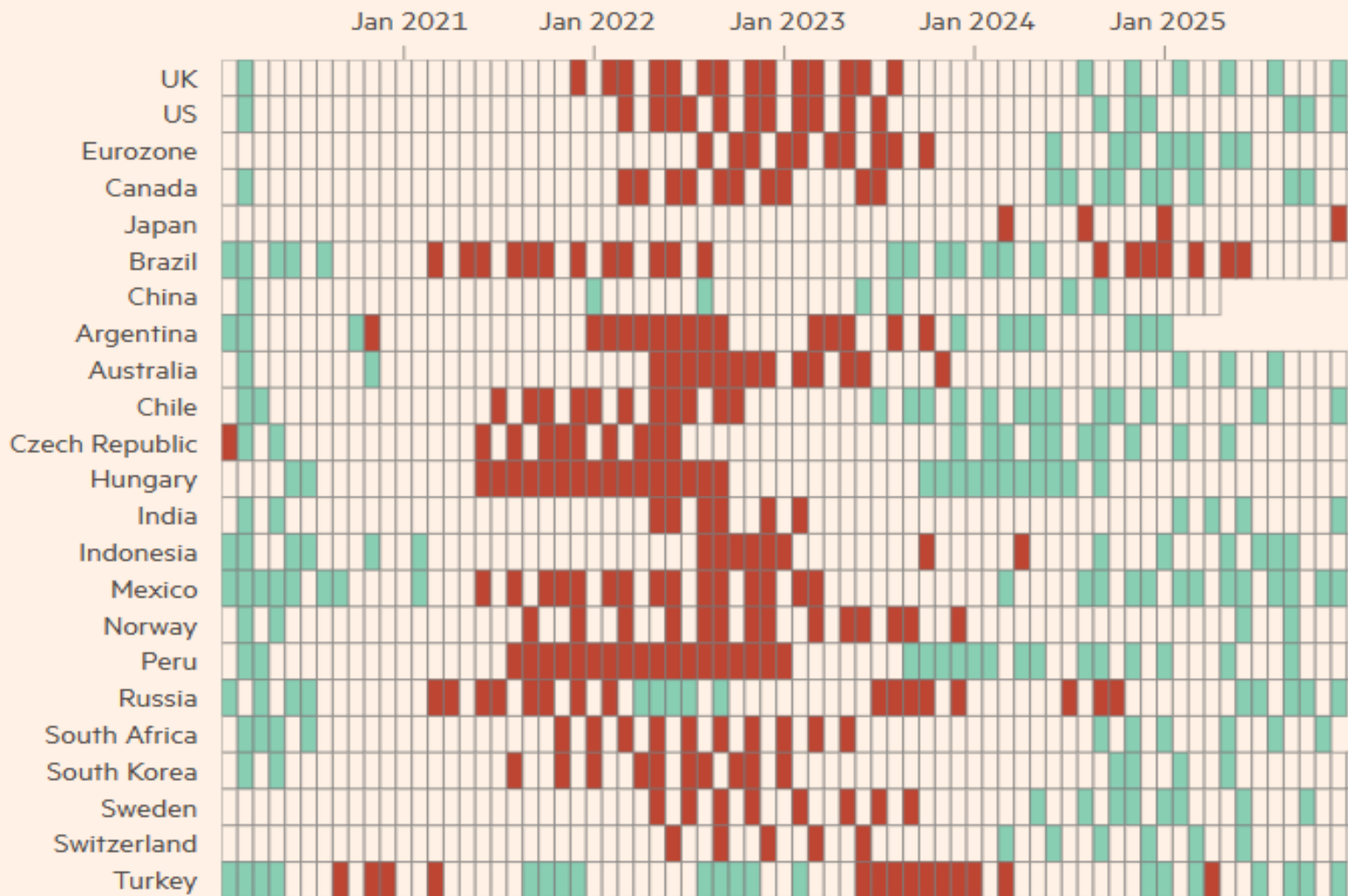
Source: FT's [inflation and interest rate tracker](#)

Central bank policy rate (%)

Choose countries...



Policy rate movement by calendar month: ■ Decrease □ No change ■ Increase



Source: FT analysis of data compiled by LSEG

Interest rate expectations

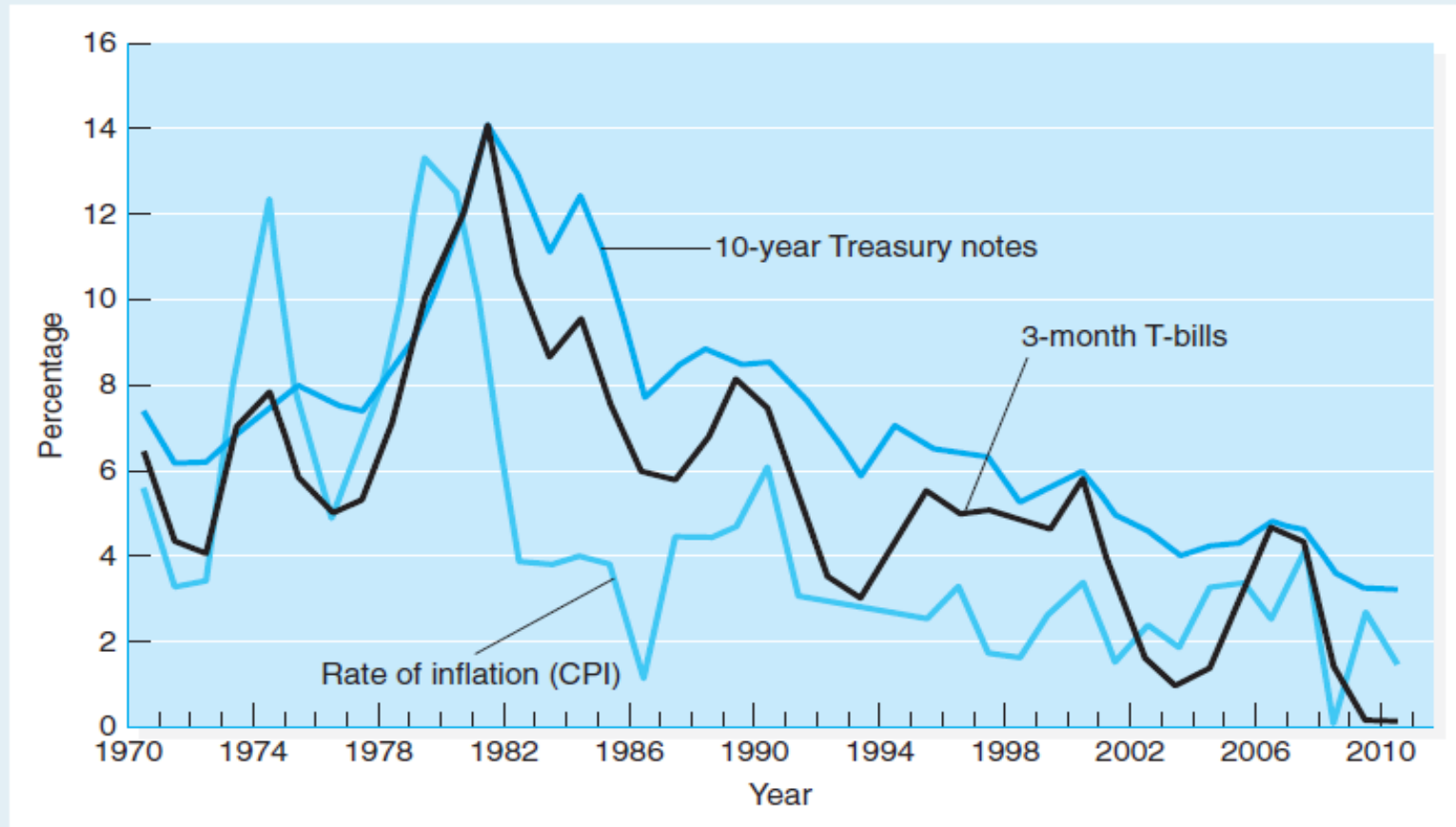
The yields on 2-year government bonds are strongly affected by market expectations of interest rates over that time.

2-year government bond yield (%)

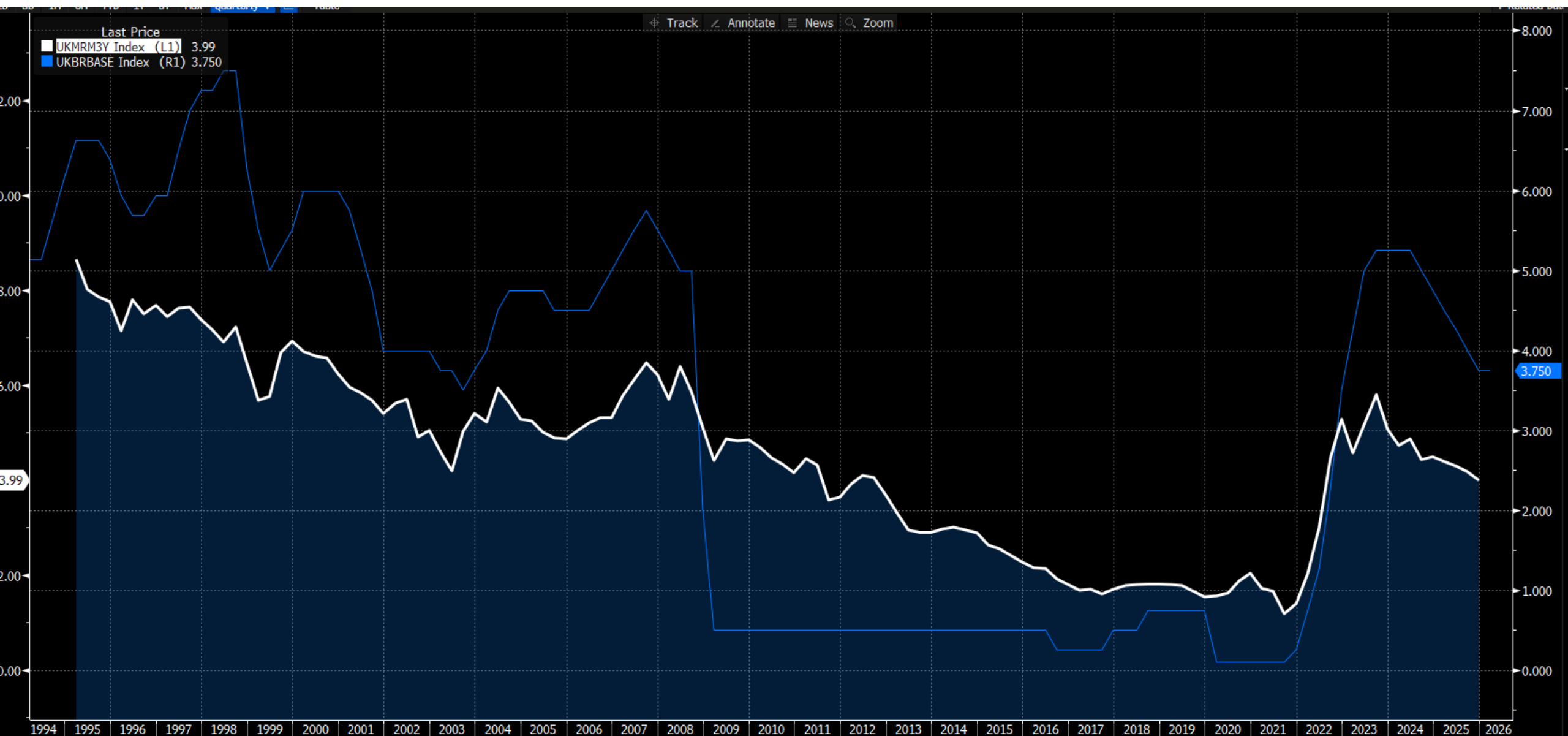


EXHIBIT 4.6

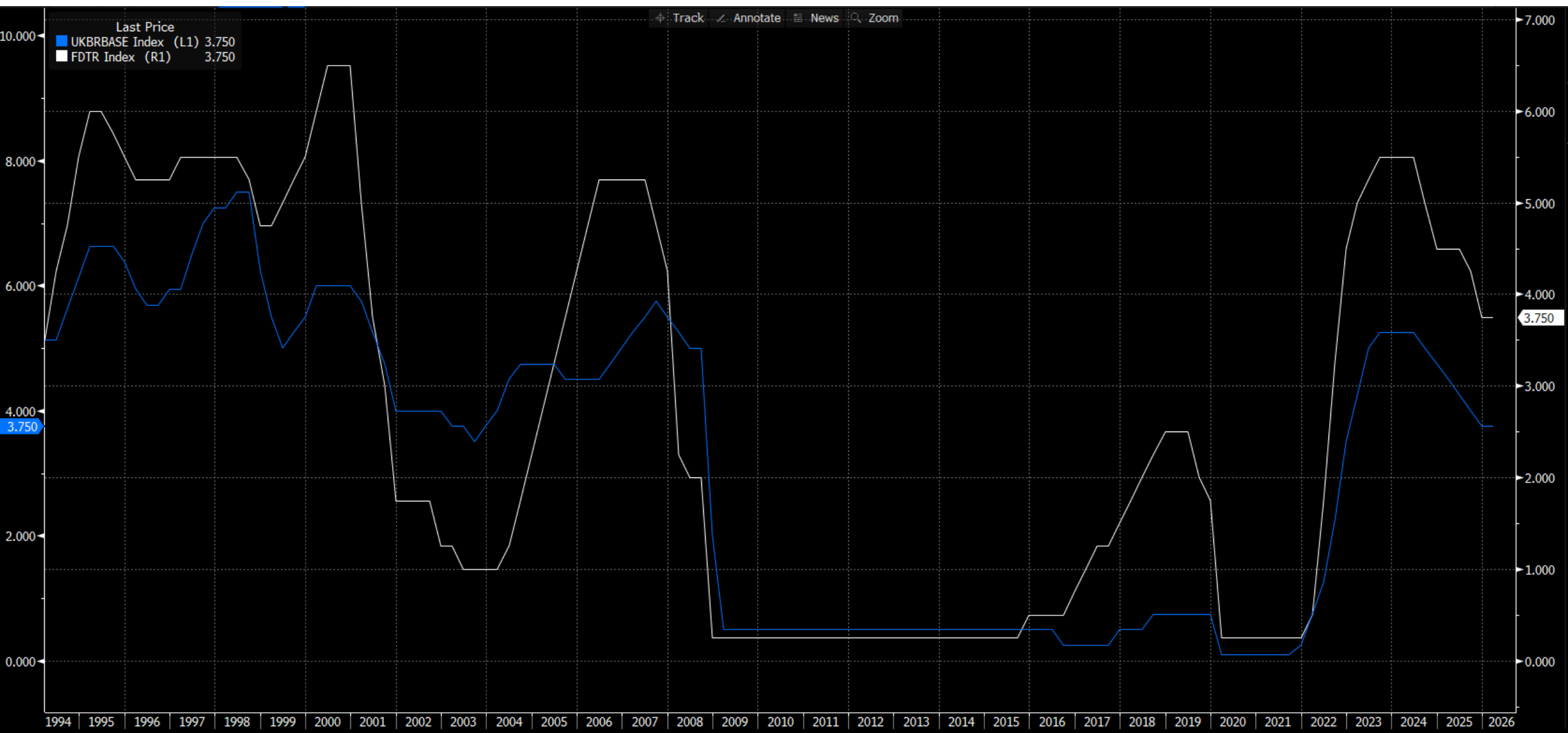
Movements of Interest Rates and Inflation (1970–2010)



Interest rates change in response to inflation, as predicted by the Fisher equation. The rate of inflation is calculated based on the Consumer Price Index (CPI).



**Source: Bloomberg,
BOE Policy rate. 2 year fixed rates on mortgages based on 75% value-to-loan as per 30-1-2026.**



General and specific rates of interest

- For simplicity, we often talk about the interest rate as though it was a single entity.
- In fact, there are lots of different interest rates, with different lenders charging different rates for different loan arrangements.
- We need, therefore, to distinguish between the general and the specific rate of interest.
- In effect, the general rate is just the average of all the prevailing specific rates.

4-Determining the General Rate of Interest

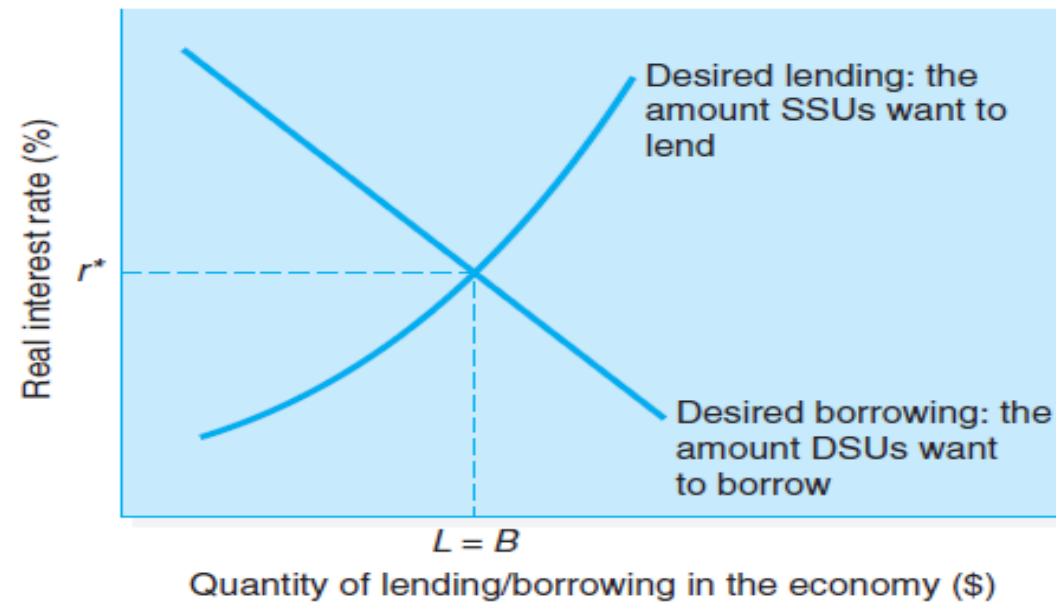
The Loanable Funds Model

- A simple model of interest rate determination which says that the general rate of interest is the rate that equates borrowers' demand for loans with lenders' willingness to supply those loans.

Determinants of the Real Rate of Interest

EXHIBIT 4.1

Determinants of the Real Interest Rate



The real rate of interest is the base interest rate for the economy. It is closely tied to the productivity of capital investments in the economy and people's time preference for consumption. The equilibrium rate of interest is the point where the desired level of borrowing (B) equals the desired level of lending (L). The real rate of interest historically has been around 3 percent for the U.S. economy and has varied between 2 and 4 percent.

Loanable Funds Theory

- Supply of loanable funds—
 - All sources of funds available to invest in financial claims
- Demand for loanable funds—
 - All uses of funds raised from issuing financial claims
- Equilibrium interest rate

Supply of loanable funds—

- All sources of funds available to invest in financial claims:
 - Consumer savings
 - Business savings
 - Government budget surpluses
 - Central Bank action

Demand for Loanable Funds

- All uses of funds raised from issuing financial claims:
 - Consumer credit purchases
 - Business investment
 - Government budget deficits

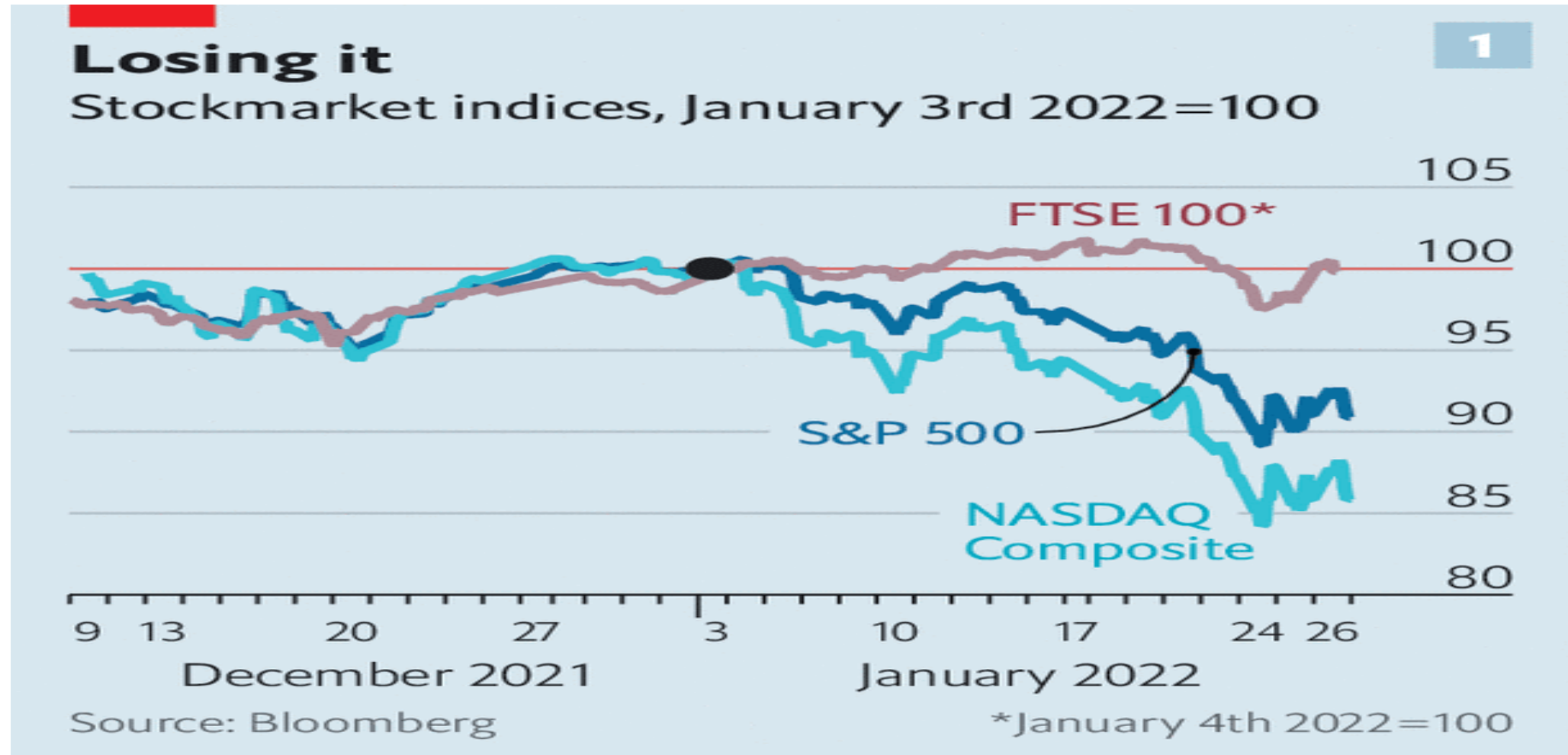
Equilibrium Interest Rate

- If competitive forces operate in financial sector, laws of supply and demand will bring rates into equilibrium.
- Equilibrium is temporary or dynamic: Any force that shifts supply or demand will tend to change interest rates.

Determining Specific Rates of Interest

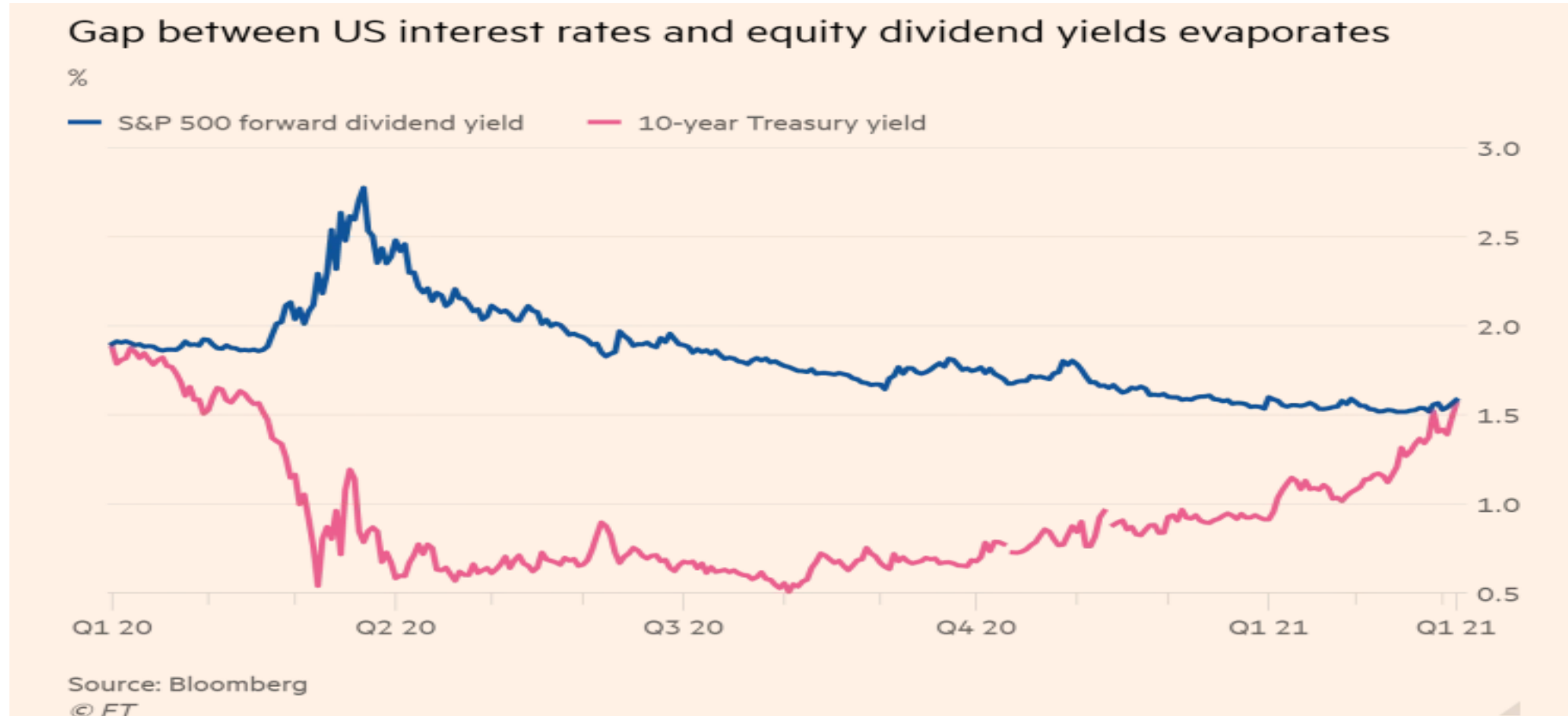
- Explanations for why the interest rate varies between borrowers
- Credit worthiness and default risk
- Duration and liquidity risk
- Secured or unsecured loans
- Differences in lenders' operating costs

5-Can changes in interest rates explain disturbances in the stock market?



The Economist

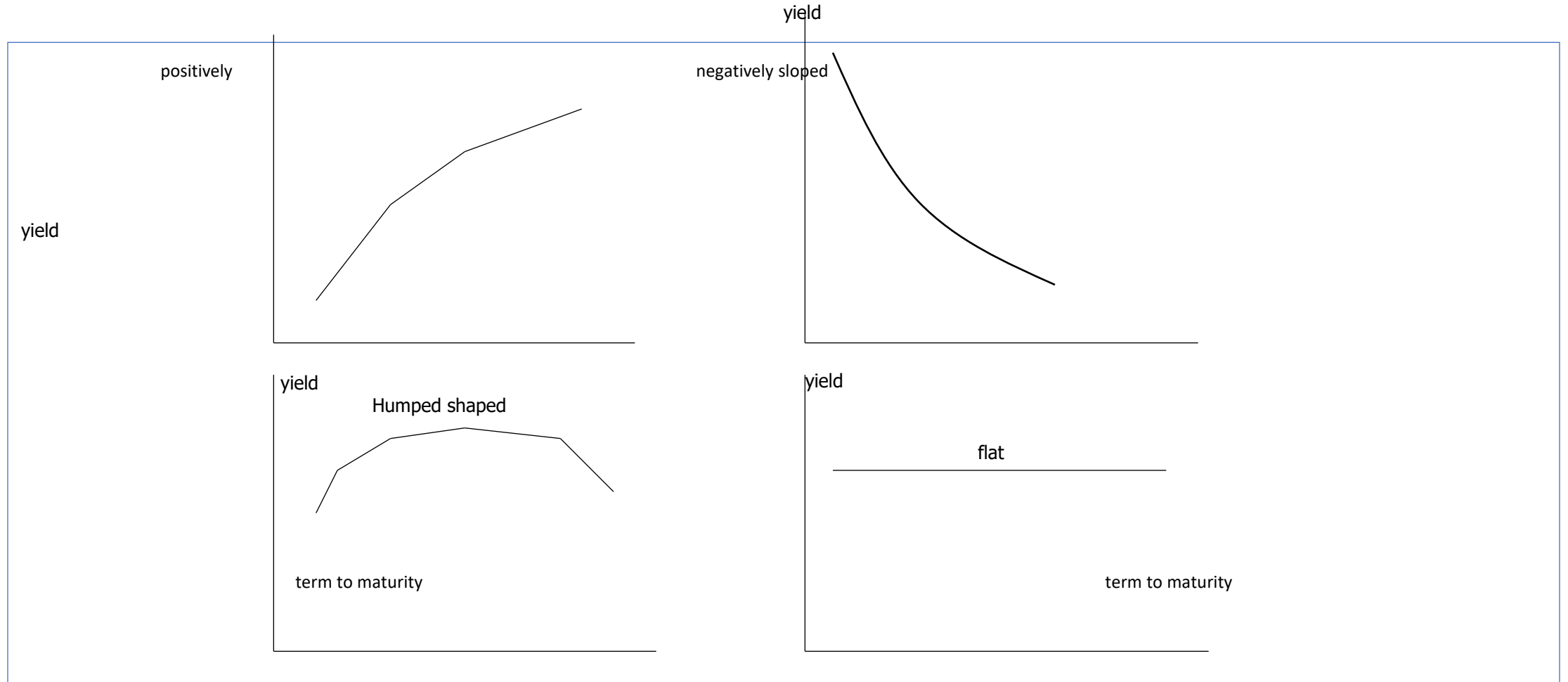
Can changes in interest rates explain disturbances in the stock market?



6-The Term Structure of Interest Rates

- The rate of interest varies between loans according to their durations.
- Typically, the longer the loan duration the higher the rate of interest charged.
- This is because the risks associated with lending increase with time, so the rate of interest must also increase.
- The relationship between loan duration and rate of interest is described by the yield curve.
- <https://www.bloomberg.com/markets/rates-bonds/government-bonds/uk>

The interest rate yield curve (Pilbeam, p91)



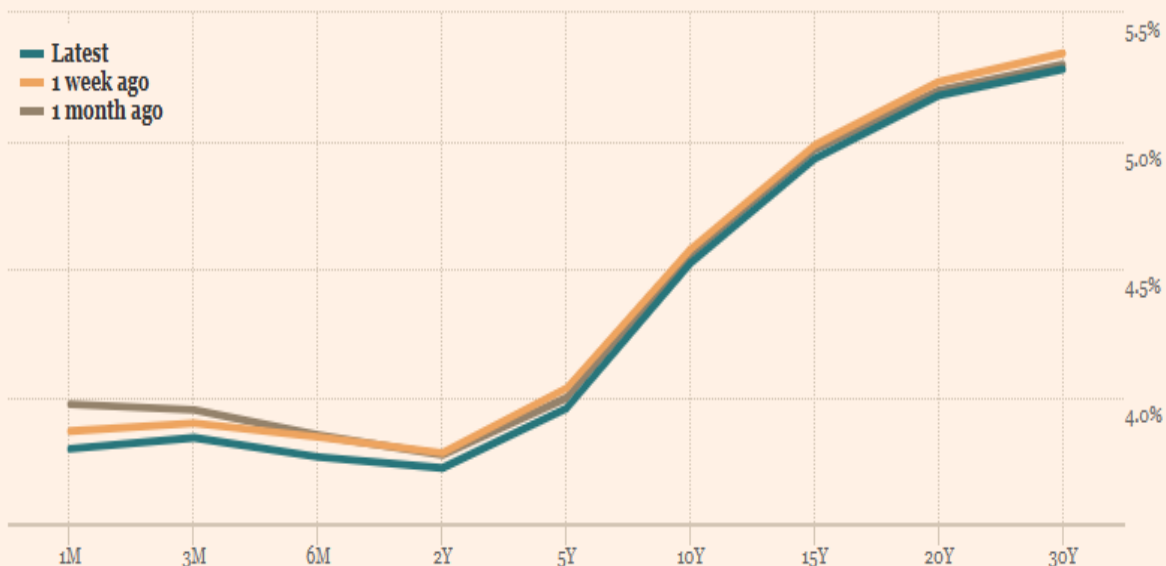
Yields

United Kingdom yield curve

United Kingdom

Chart

Table



Yields

United Kingdom

Chart

Table

Maturity	Yield	Today's change	1 week ago	1 month ago
1 Month	3.80%	<0.01	3.87%	3.97%
3 Month	3.84%	0.00	3.90%	3.95%
6 Month	3.77%	<0.01	3.84%	3.85%
1 Year	3.59%	<0.01	3.77%	3.63%
2 Year	3.72%	<0.01	3.78%	3.77%
3 Year	3.78%	>-0.01	3.85%	3.82%
4 Year	3.88%	<0.01	3.96%	3.93%
5 Year	3.96%	<0.01	4.03%	4.00%
7 Year	4.12%	<0.01	4.18%	4.15%
8 Year	4.40%	+0.01	4.46%	4.43%
9 Year	4.45%	+0.01	4.51%	4.49%
10 Year	4.52%	+0.01	4.58%	4.56%
15 Year	4.93%	+0.01	4.98%	4.97%
20 Year	5.18%	+0.01	5.23%	5.20%
30 Year	5.28%	+0.01	5.34%	5.30%

Data delayed at least 15 minutes, as of Jan 30 2026 09:29 GMT.

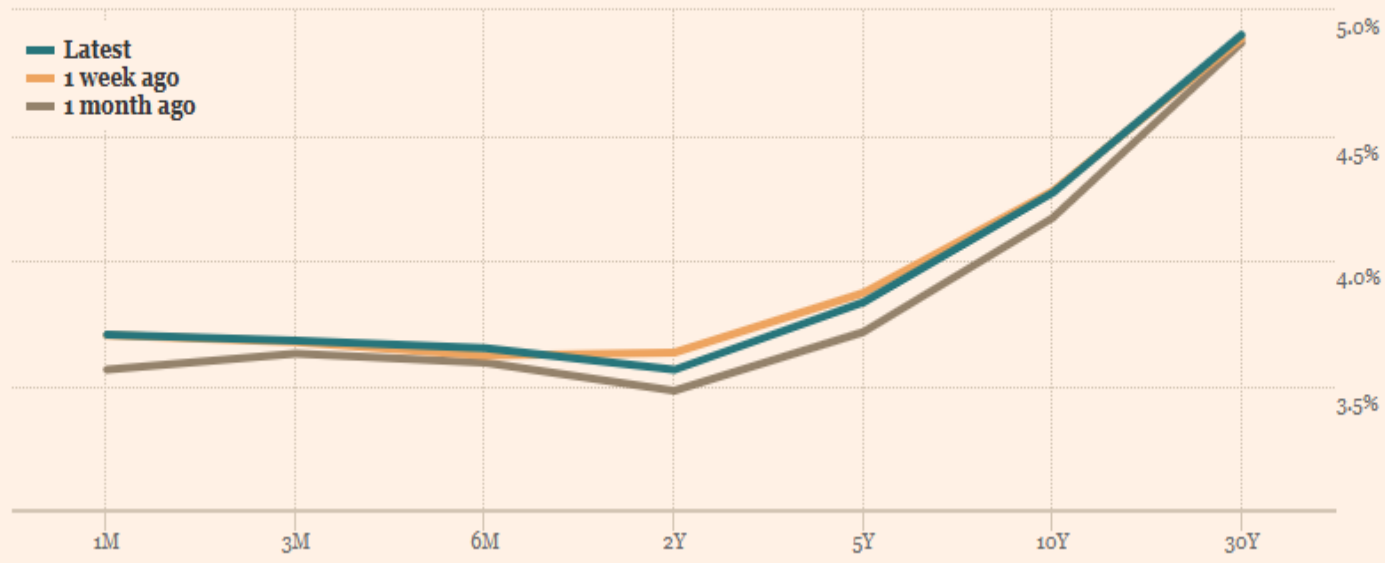
Source: <https://markets.ft.com/data/bonds>

As per 30-1-2026

Yields

United States yield curve

United States Chart Table



Yields

United States Chart Table

Maturity	Yield	Today's change	1 week ago	1 month ago
1 Month	3.70%	+0.01	3.70%	3.56%
3 Month	3.68%	+0.01	3.67%	3.63%
6 Month	3.65%	<0.01	3.62%	3.59%
2 Year	3.56%	+0.01	3.63%	3.48%
5 Year	3.83%	+0.03	3.87%	3.71%
10 Year	4.27%	+0.04	4.28%	4.17%
30 Year	4.90%	+0.05	4.89%	4.87%

Data delayed at least 15 minutes, as of Jan 30 2026 09:29 GMT.

Source: <https://markets.ft.com/data/bonds>
As per 30-1-2026

Ten year government bond spreads

Country	Latest yield	Spread vs bund	Spread vs T-notes
Australia	4.79%	+1.94	+0.48
Austria	3.05%	+0.19	-1.27
Belgium	3.26%	+0.41	-1.05
Canada	3.43%	+0.57	-0.89
Denmark	2.72%	-0.13	-1.60
Finland	3.11%	+0.26	-1.20
France	3.44%	+0.58	-0.88
Germany	2.85%	--	-1.46
Greece	3.36%	+0.51	-0.96
Ireland	2.96%	+0.11	-1.35
Italy	3.47%	+0.62	-0.84
Japan	2.25%	-0.60	-2.06
Netherlands	2.92%	+0.07	-1.39
New Zealand	4.64%	+1.79	+0.32
Portugal	3.21%	+0.36	-1.11
Spain	3.22%	+0.37	-1.09
Sweden	2.76%	-0.09	-1.55
Switzerland	0.23%	-2.62	-4.09
UK	4.52%	+1.67	+0.21
US	4.27%	+1.42	--

Data delayed at least 15 minutes, as of Jan 30 2026 11:42 GMT.

Source: <https://markets.ft.com/data/bonds>

As per 30-1-2026

7-Expectations and the term structure of interest rates

- Although the yield curve typically slopes up from left to right, it can slope downwards or even go up before turning down again.
- This is because the yield curve is telling us what lenders currently expect to happen in the future.
- If, for example, lenders expect inflation to fall in the long term, the yield curve will rise over the short and medium terms, but turn down in the long term